

1. IFC4.3 delivery

```
#103=IFCPLATE('2LiDpsq0XSP0GZ$amUn_RZ..  
#136=IFCMECHANICALFASTENER('1NsEaARyr...
```

2. Scene IR / CEWG

```
"id": "plate_pin_insertion"  
"component": "connection_plate"  
"mate_component": "assembly_pin"  
"origin_m": [2.11, 1.5, 1.0]  
"task": "insert_pin"  
"target": "connection_plate"
```

3. OpenUSD composition

```
#usda 1.0  building_root.usda  
csr:compositionMode="task_activated"  
root: building_root, all_loaded  
zones: zone_a, zone_b, zone_c
```

4. Scene validator

```
rule_id: CSR-SCN-028  
severity: critical  
path: $.tasks[0].interface  
message: The task target shall own the  
         referenced assembly interface.
```

5. Whitelisted repair

```
tool: restore_interface_target_binding  
action: restore target_component  
        from evidence record  
benchmark outcome: valid
```

6. Runtime measurement

```
R-REPAIR-GE0-0001 (200 eps each)  
correct:   reward 87.0%  geo 43.5%  
fault(5mm): reward 84.0%  geo 22.5%  
repaired:  reward 88.5%  geo 34.0%
```